

BuzzCalculus 作業本

Calculus Reflex Training Workbook

總題數	426 題
題庫分類	
極限	61 題
微分與多變數	148 題
積分	148 題
級數與收斂	69 題

姓名：_____

日期：_____

分數：_____

本作業本由 BuzzCalculus 題庫自動產生。建議用 XeLaTeX + XeCJK 編譯。

Contents

1	極限 Limits	2
2	微分與多變數 Derivatives	12
3	積分 Integrals	38
4	級數與收斂 Series	70
	答案索引	82

1 極限 Limits

共 61 題。題目依 BuzzCalculus 題庫順序排列，適合列印後作為練習本使用。

1 01 / lim-001

極限 / Rank 1/6 Warm-up / 數值 / rank-1 / beginner-friendly

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

2 02 / lim-002

極限 / Rank 1/6 Warm-up / 數值 / rank-1 / beginner-friendly

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

3 03 / lim-003

極限 / Rank 1/6 Warm-up / 數值 / rank-1 / beginner-friendly

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

4 04 / lim-004

極限 / Rank 1/6 Warm-up / 數值 / rank-1 / beginner-friendly

$$\lim_{x \rightarrow \infty} \frac{3x^2 - x + 7}{2x^2 + 5}$$

5 05 / lim-005

極限 / Rank 2/6 Basic / 數值 / rank-2 / beginner-friendly

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$$

6 06 / lim-006

極限 / Rank 2/6 Basic / 數值 / rank-2 / beginner-friendly

$$\lim_{x \rightarrow 0} \frac{\ln(1 + 3x)}{x}$$

7 07 / lim-007

極限 / Rank 2/6 Basic / 數值 / rank-2 / beginner-friendly

$$\lim_{x \rightarrow 0} \frac{\tan(4x)}{\sin(2x)}$$

8 08 / lim-008

極限 / Rank 2/6 Basic / 數值 / rank-2 / beginner-friendly

$$\lim_{x \rightarrow \infty} x \left(\sqrt{x^2 + 1} - x \right)$$

9 09 / lim-009

極限 / Rank 3/6 Standard / 數值 / rank-3

$$\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$$

10 10 / lim-010

極限 / Rank 3/6 Standard / 數值 / rank-3

$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2}$$

11 11 / lim-011

極限 / Rank 3/6 Standard / 數值 / rank-3

$$\lim_{x \rightarrow 0^+} x \ln x$$

12 12 / lim-012

極限 / Rank 3/6 Standard / 數值 / rank-3

$$\lim_{x \rightarrow 0} \frac{\ln(1+x) - x + \frac{x^2}{2}}{x^3}$$

13 13 / lim-013

極限 / Rank 3/6 Standard / 數值 / rank-3

$$\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$$

14 14 / lim-014

極限 / Rank 4/6 Advanced / 數值 / rank-4

$$\lim_{x \rightarrow 0} \frac{\ln(1+x) - \sin x}{x^2}$$

15 15 / lim-015

極限 / Rank 3/6 Standard / 數值 / rank-3

$$\lim_{x \rightarrow 0} \frac{\sec x - 1}{x^2}$$

16 16 / lim-016

極限 / Rank 4/6 Advanced / 數值 / rank-4

$$\lim_{x \rightarrow \infty} x \left(\sqrt{x^2 + x} - x - \frac{1}{2} \right)$$

17 17 / lim-017

極限 / Rank 3/6 Standard / 數值 / rank-3

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x^3}$$

18 18 / lim-018

極限 / Rank 3/6 Standard / 數值 / rank-3

$$\lim_{x \rightarrow 0} \frac{\sin x - x \cos x}{x^3}$$

19 19 / lim-019

極限 / Rank 4/6 Advanced / 數值 / rank-4

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e}{x}$$

20 20 / lim-020

極限 / Rank 3/6 Standard / 數值 / rank-3

$$\lim_{x \rightarrow 0} \frac{\arctan x - x}{x^3}$$

21 21 / td-lim-001

極限 / Rank 6/6 Boss+ / 數值 / rank-6 / boss-rank / boss-plus

$$\lim_{n \rightarrow \infty} \prod_{k=1}^n \left(1 + \frac{k}{n^2}\right)$$

22 22 / td-lim-002

極限 / Rank 6/6 Boss+ / 數值 / rank-6 / boss-rank / boss-plus

$$\lim_{n \rightarrow \infty} n \left(\prod_{k=1}^n \left(1 + \frac{k}{n^2}\right) - \sqrt{e} \right)$$

23 23 / td-lim-003

極限 / Rank 6/6 Boss+ / 數值 / rank-6 / boss-rank / boss-plus

$$\lim_{n \rightarrow \infty} n^2 \left(\left(1 + \frac{1}{n}\right)^n - e + \frac{e}{2n} \right)$$

24 24 / td-lim-004

極限 / Rank 4/6 Advanced / 數值 / rank-4

$$\lim_{x \rightarrow 0} \frac{(1+x)^x - 1 - x^2}{x^3}$$

25 25 / lim-021

極限 / Rank 2/6 Basic / 數值 / taylor / trig-limit / rank-2

$$\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$$

26 26 / lim-022

極限 / Rank 2/6 Basic / 數值 / rationalize / infinity / rank-2

$$\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 3x} - x \right)$$

27 27 / lim-023

極限 / Rank 2/6 Basic / 數值 / taylor / exponential / rank-2

$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2}$$

28 28 / lim-024

極限 / Rank 2/6 Basic / 數值 / taylor / log / rank-2

$$\lim_{x \rightarrow 0} \frac{\log(1+x) - x}{x^2}$$

29 29 / lim-025

極限 / Rank 4/6 Advanced / 數值 / multivariable / standard-limit / rank-4

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(xy)}{xy}$$

30 30 / lim-026

極限 / Rank 4/6 Advanced / 數值 / multivariable / squeeze / rank-4

$$\lim_{(x,y)\rightarrow(0,0)} \frac{x^2y}{x^2+y^2}$$

31 31 / lim-027

極限 / Rank 4/6 Advanced / 數值 / multivariable / path-test / dne

$$\lim_{(x,y)\rightarrow(0,0)} \frac{xy}{x^2+y^2}$$

32 32 / lim-028

極限 / Rank 4/6 Advanced / 數值 / multivariable / rationalize / rank-4

$$\lim_{(x,y)\rightarrow(0,0)} \frac{\sqrt{1+x^2+y^2}-1}{x^2+y^2}$$

33 33 / lim-029

極限 / Rank 4/6 Advanced / 數值 / multivariable / squeeze / rank-4

$$\lim_{(x,y)\rightarrow(0,0)} \frac{x^4+y^4}{x^2+y^2}$$

34 34 / lim-030

極限 / Rank 2/6 Basic / 數值 / standard-limit / trig-limit / rank-2

$$\lim_{x\rightarrow 0} \frac{1-\cos(3x)}{x^2}$$

35 35 / lim-031

極限 / Rank 4/6 Advanced / 數值 / multivariable / squeeze / rank-4

$$\lim_{(x,y)\rightarrow(0,0)} \frac{x^2y^2}{x^2+y^2}$$

36 36 / lim-032

極限 / Rank 4/6 Advanced / 數值 / multivariable / path-test / dne

$$\lim_{(x,y)\rightarrow(0,0)} \frac{x^2-y^2}{x^2+y^2}$$

37 37 / lim-033

極限 / Rank 4/6 Advanced / 數值 / multivariable / standard-limit / exponential

$$\lim_{(x,y)\rightarrow(0,0)} \frac{e^{x+y}-1}{x+y}$$

38 38 / lim-034

極限 / Rank 4/6 Advanced / 數值 / multivariable / squeeze / rank-4

$$\lim_{(x,y)\rightarrow(0,0)} \frac{x^3}{x^2+y^2}$$

39 39 / lim-035

極限 / Rank 2/6 Basic / 數值 / standard-limit / trig-limit / rank-2

$$\lim_{x\rightarrow 0} \frac{\sin(5x)}{\sin(2x)}$$

40 40 / lim-036

極限 / Rank 3/6 Standard / 數值 / taylor / log / rank-3

$$\lim_{x\rightarrow 0} \frac{\log(1+x+x^2)-x}{x^2}$$

41 41 / lim-037

極限 / Rank 2/6 Basic / 數值 / rationalize / rank-2 / beginner-friendly

$$\lim_{x\rightarrow 0} \frac{\sqrt{1+4x}-1}{x}$$

42 42 / lim-038

極限 / Rank 3/6 Standard / 數值 / taylor / trig-limit / rank-3

$$\lim_{x \rightarrow 0} \frac{\sin x - x \cos x}{x^3}$$

43 43 / lim-039

極限 / Rank 4/6 Advanced / 數值 / multivariable / path-test / dne

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$$

44 44 / lim-040

極限 / Rank 2/6 Basic / 數值 / standard-limit / log / infinity

$$\lim_{x \rightarrow \infty} x \log \left(1 + \frac{2}{x} \right)$$

45 45 / gap-tech-001

極限 / Rank 2/6 Basic / 文字 / technique-recognition / rationalize / rank-2

Technique for $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 3x} - x \right)$

46 46 / gap-tech-002

極限 / Rank 2/6 Basic / 文字 / technique-recognition / taylor / rank-2

Technique for $\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2}$

47 47 / gap-tech-008

極限 / Rank 2/6 Basic / 文字 / technique-recognition / multivariable-limit / path-test

Technique for $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$

48 48 / mob-tech-001

極限 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$\lim_{x \rightarrow \infty} (\sqrt{x^2 + 3x} - x)$ Use which technique?

49 49 / mob-tech-002

極限 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2}$ Use which technique?

50 50 / mob-tech-009

極限 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$ Use which technique?

51 51 / mob-trap-009

極限 / Rank 2/6 Basic / 文字 / trap-drill / rank-2 / beginner-friendly

$\frac{0}{0}$ limit form Main trap?

52 52 / mob-limtrap-001

極限 / Rank 3/6 Standard / 數值 / limit-trap / taylor / rank-3

$\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$

53 53 / mob-limtrap-002

極限 / Rank 3/6 Standard / 數值 / limit-trap / taylor / rank-3

$\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$

54 54 / mob-limtrap-003

極限 / Rank 3/6 Standard / 數值 / limit-trap / taylor / rank-3

$$\lim_{x \rightarrow 0} \frac{1 - \cos x - x^2/2}{x^4}$$

55 55 / mob-limtrap-004

極限 / Rank 3/6 Standard / 數值 / limit-trap / taylor / rank-3

$$\lim_{x \rightarrow 0} \frac{\log(1+x) - x + x^2/2}{x^3}$$

56 56 / mob-limtrap-005

極限 / Rank 3/6 Standard / 數值 / limit-trap / taylor / rank-3

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x - x^2/2}{x^3}$$

57 57 / mob-limtrap-006

極限 / Rank 3/6 Standard / 數值 / limit-trap / taylor / rank-3

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1 - x/2}{x^2}$$

58 58 / mob-limtrap-007

極限 / Rank 3/6 Standard / 數值 / limit-trap / taylor / rank-3

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 + y^2}$$

59 59 / mob-limtrap-008

極限 / Rank 3/6 Standard / 數值 / limit-trap / taylor / rank-3

$$\lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2 + y^2} \sin \frac{1}{\sqrt{x^2 + y^2}}$$

60 60 / mob-limtrap-009

極限 / Rank 3/6 Standard / 文字 / limit-trap / path-test / rank-3

$$\lim_{(x,y)\rightarrow(0,0)} \frac{xy}{x^2+y^2}$$

61 61 / mob-limtrap-010

極限 / Rank 3/6 Standard / 文字 / limit-trap / path-test / rank-3

$$\lim_{(x,y)\rightarrow(0,0)} \frac{x^2-y^2}{x^2+y^2}$$

2 微分與多變數 Derivatives

共 148 題。題目依 BuzzCalculus 題庫順序排列，適合列印後作為練習本使用。

62 01 / der-001

微分與多變數 / Rank 1/6 Warm-up / 表達式 / rank-1 / beginner-friendly

$$\frac{d}{dx} (x^5 - 3x^2 + 7)$$

63 02 / der-002

微分與多變數 / Rank 1/6 Warm-up / 表達式 / rank-1 / beginner-friendly

$$\frac{d}{dx} (\sin x + \cos x)$$

64 03 / der-003

微分與多變數 / Rank 1/6 Warm-up / 表達式 / rank-1 / beginner-friendly

$$\frac{d}{dx} (e^x \ln x)$$

65 04 / der-004

微分與多變數 / Rank 1/6 Warm-up / 表達式 / rank-1 / beginner-friendly

$$\frac{d}{dx} (\sqrt{x})$$

66 05 / der-005

微分與多變數 / Rank 2/6 Basic / 表達式 / rank-2 / beginner-friendly

$$\frac{d}{dx}(\ln(1+x^2))$$

67 06 / der-006

微分與多變數 / Rank 2/6 Basic / 表達式 / rank-2 / beginner-friendly

$$\frac{d}{dx}(x^2 \sin x)$$

68 07 / der-007

微分與多變數 / Rank 2/6 Basic / 表達式 / rank-2 / beginner-friendly

$$\frac{d}{dx}(\arctan x)$$

69 08 / der-008

微分與多變數 / Rank 2/6 Basic / 表達式 / rank-2 / beginner-friendly

$$\frac{d}{dx}\left(\frac{x}{x+1}\right)$$

70 09 / der-009

微分與多變數 / Rank 3/6 Standard / 表達式 / rank-3

$$\frac{d}{dx}(x^x)$$

71 10 / der-010

微分與多變數 / Rank 3/6 Standard / 表達式 / rank-3

$$\frac{d}{dx}(\sin(x^2)e^{-x})$$

72 11 / der-011

微分與多變數 / Rank 3/6 Standard / 表達式 / rank-3

$$\frac{d}{dx}(\ln(\sin x))$$

73 12 / der-012

微分與多變數 / Rank 4/6 Advanced / 表達式 / rank-4

$$\frac{d}{dx}\left(\frac{\ln x}{x^2 + 1}\right)$$

74 13 / der-013

微分與多變數 / Rank 4/6 Advanced / 表達式 / rank-4

$$\frac{d}{dx}(x^{\sin x})$$

75 14 / der-014

微分與多變數 / Rank 3/6 Standard / 表達式 / rank-3

$$\frac{d}{dx}\left(\arctan \frac{1+x}{1-x}\right)$$

76 15 / der-015

微分與多變數 / Rank 3/6 Standard / 表達式 / rank-3

$$\frac{d}{dx}\left(\ln\left(x + \sqrt{1+x^2}\right)\right)$$

77 16 / der-016

微分與多變數 / Rank 3/6 Standard / 表達式 / rank-3

$$\frac{d}{dx}\left(\frac{\sin x}{1 + \cos x}\right)$$

78 17 / der-017

微分與多變數 / Rank 3/6 Standard / 表達式 / rank-3

$$\frac{d}{dx}\left(e^{x^2}\cos(3x)\right)$$

79 18 / der-018

微分與多變數 / Rank 3/6 Standard / 表達式 / rank-3

$$\frac{d}{dx}\left(\ln\left(x+\sqrt{x^2-1}\right)\right)$$

80 19 / der-019

微分與多變數 / Rank 2/6 Basic / 表達式 / rank-2 / beginner-friendly

$$\frac{d}{dx}\left((x^2+1)^5\right)$$

81 20 / der-020

微分與多變數 / Rank 3/6 Standard / 表達式 / rank-3

$$\frac{d}{dx}\left(\arctan\sqrt{x}\right)$$

82 21 / td-der-001

微分與多變數 / Rank 4/6 Advanced / 表達式 / rank-4

$$\frac{d^3}{dx^3}\arctan\frac{1}{x}$$

83 22 / td-der-002

微分與多變數 / Rank 2/6 Basic / 表達式 / rank-2 / beginner-friendly

$$\frac{d}{dx}\left[\frac{1}{8}(35x^4-30x^2+3)\right]$$

84 23 / td-der-003

微分與多變數 / Rank 3/6 Standard / 表達式 / rank-3

$$\frac{d}{dx} (x \log(1+x))$$

85 24 / td-der-004

微分與多變數 / Rank 3/6 Standard / 表達式 / rank-3

$$\frac{d}{dx} (\log(5 + 3 \cos x))$$

86 25 / der-021

微分與多變數 / Rank 4/6 Advanced / 表達式 / multivariable / partial-derivative / chain-rule

$$\frac{\partial}{\partial x} (x^2 y + \sin(xy))$$

87 26 / der-022

微分與多變數 / Rank 4/6 Advanced / 表達式 / multivariable / partial-derivative / chain-rule

$$\frac{\partial}{\partial y} (xe^{xy} + y^3)$$

88 27 / der-023

微分與多變數 / Rank 4/6 Advanced / 表達式 / multivariable / mixed-partial / chain-rule

$$\frac{\partial^2}{\partial x \partial y} e^{xy}$$

89 28 / der-024

微分與多變數 / Rank 4/6 Advanced / 數值 / multivariable / directional-derivative / gradient

Directional derivative of $f = x^2 + y^2$ at $(1, 2)$ in direction $(3, 4)/5$

90 29 / der-025

微分與多變數 / Rank 4/6 Advanced / 表達式 / multivariable / partial-derivative / log

$$\frac{\partial}{\partial x} \log(x^2 + y^2)$$

91 30 / der-026

微分與多變數 / Rank 2/6 Basic / 表達式 / product-rule / chain-rule / rank-2

$$\frac{d}{dx} (x^2 \sin(3x))$$

92 31 / der-027

微分與多變數 / Rank 2/6 Basic / 表達式 / log / chain-rule / rank-2

$$\frac{d}{dx} \log \sqrt{1 + x^2}$$

93 32 / der-028

微分與多變數 / Rank 2/6 Basic / 表達式 / inverse-trig / chain-rule / rank-2

$$\frac{d}{dx} \arctan(e^x)$$

94 33 / der-029

微分與多變數 / Rank 2/6 Basic / 表達式 / second-derivative / product-rule / rank-2

$$\frac{d^2}{dx^2} (xe^x)$$

95 34 / der-030

微分與多變數 / Rank 3/6 Standard / 表達式 / implicit-differentiation / rank-3

Find y' from $x^2 + xy + y^2 = 7$

96 35 / der-031

微分與多變數 / Rank 4/6 Advanced / 表達式 / multivariable / partial-derivative / chain-rule

$$\frac{\partial}{\partial x} \left(e^{x^2+y^2} \right)$$

97 36 / der-032

微分與多變數 / Rank 4/6 Advanced / 表達式 / multivariable / partial-derivative / inverse-trig

$$\frac{\partial}{\partial y} \arctan \left(\frac{y}{x} \right)$$

98 37 / der-033

微分與多變數 / Rank 4/6 Advanced / 表達式 / multivariable / second-derivative / rank-4

$$\frac{\partial^2}{\partial x^2} (x^2y + \sin x)$$

99 38 / der-034

微分與多變數 / Rank 4/6 Advanced / 表達式 / multivariable / jacobian / rank-4

Jacobian determinant of $u = x + y, v = xy$

100 39 / der-035

微分與多變數 / Rank 4/6 Advanced / 表達式 / multivariable / chain-rule / rank-4

If $z = x^2y, x = t^2, y = e^t$, find dz/dt

101 40 / der-036

微分與多變數 / Rank 4/6 Advanced / 數值 / multivariable / second-derivative / laplacian

Laplacian of $f = x^2 + y^2$

102 41 / der-037

微分與多變數 / Rank 4/6 Advanced / 數值 / multivariable / hessian / second-derivative

Hessian determinant of $f = x^3 + y^3 - 3xy$ at $(1, 1)$

103 42 / der-038

微分與多變數 / Rank 2/6 Basic / 表達式 / quotient-rule / rank-2 / beginner-friendly

$$\frac{d}{dx} \left(\frac{x^2 + 1}{x - 1} \right)$$

104 43 / der-039

微分與多變數 / Rank 2/6 Basic / 表達式 / chain-rule / trig / rank-2

$$\frac{d}{dx} (\sin^2 x)$$

105 44 / der-040

微分與多變數 / Rank 3/6 Standard / 表達式 / logarithmic-differentiation / rank-3

$$\frac{d}{dx} (x^{\log x})$$

106 45 / der-041

微分與多變數 / Rank 4/6 Advanced / 數值 / total-differential / linearization / estimate

Use total differential to estimate Δf for $f = x^2y$ at $(2, 1)$, $dx = 0.01$, $dy = -0.02$

107 46 / der-042

微分與多變數 / Rank 4/6 Advanced / 數值 / total-differential / linearization / estimate

Use total differential for $f = \sqrt{x^2 + y^2}$ at $(3, 4)$, $dx = 0.06$, $dy = -0.08$

108 47 / der-043

微分與多變數 / Rank 4/6 Advanced / 數值 / total-differential / linearization / estimate

Estimate Δz for $z = \log(xy)$ at $(1, 2)$, $dx = 0.02$, $dy = -0.04$

109 48 / der-044

微分與多變數 / Rank 4/6 Advanced / 表達式 / total-differential / linearization / rank-4

Linear part df of $f = e^{xy}$

110 49 / der-045

微分與多變數 / Rank 4/6 Advanced / 表達式 / total-differential / linearization / inverse-trig

Linear part df of $f = \arctan(y/x)$

111 50 / der-046

微分與多變數 / Rank 4/6 Advanced / 數值 / total-differential / linearization / estimate

Linear estimate of $x^2 + y^2$ at $(1.02, 1.97)$ using base $(1, 2)$

112 51 / der-047

微分與多變數 / Rank 5/6 Boss / 數值 / total-differential-min / optimization / quadratic

$$\min_{x,y} (x^2 + 2xy + 3y^2 - 4x - 6y)$$

113 52 / der-048

微分與多變數 / Rank 5/6 Boss / 數值 / total-differential-min / optimization / quadratic

$$\min_{x,y} (3x^2 + 2xy + 2y^2 - 8x - 6y)$$

114 53 / der-049

微分與多變數 / Rank 5/6 Boss / 數值 / total-differential-min / optimization / quadratic

x-coordinate of the minimum of $x^2 + y^2 + 2x - 4y + 7$

115 54 / der-050

微分與多變數 / Rank 5/6 Boss / 數值 / total-differential-min / optimization / constraint

$\min (x^2 + y^2)$ subject to $x + 2y = 5$

116 55 / der-051

微分與多變數 / Rank 4/6 Advanced / 數值 / hessian / second-derivative / rank-4

Hessian determinant of $f = x^2 + xy + y^2$

117 56 / der-052

微分與多變數 / Rank 4/6 Advanced / 數值 / hessian / second-derivative / rank-4

Hessian determinant of $f = x^4 + y^4$ at $(1, 1)$

118 57 / der-053

微分與多變數 / Rank 4/6 Advanced / 數值 / hessian / second-derivative / exponential

Hessian determinant of $f = e^{xy}$ at $(0, 0)$

119 58 / der-054

微分與多變數 / Rank 4/6 Advanced / 數值 / hessian / second-derivative / rank-4

Hessian determinant of $f = x^4 + y^4 - 4xy$ at $(1, 1)$

Classify the critical point $(0, 0)$ of $f = x^2 + y^2$

Classify the critical point $(0, 0)$ of $f = x^2 - y^2$

Hessian determinant of $f = x^4 + y^2$ at $(0, 0)$

Hessian determinant of $f = \log(x^2 + y^2)$ at $(1, 0)$

Wronskian $W(\sin x, \cos x)$

Wronskian $W(\cos x, \sin x)$

126 65 / der-061

微分與多變數 / Rank 4/6 Advanced / 表達式 / wronskian / linear-independence / exponential

Wronskian $W(e^x, e^{2x})$

127 66 / der-062

微分與多變數 / Rank 4/6 Advanced / 表達式 / wronskian / polynomial / rank-4

Wronskian $W(x, x^2)$

128 67 / der-063

微分與多變數 / Rank 4/6 Advanced / 數值 / wronskian / polynomial / rank-4

Wronskian $W(1, x, x^2)$

129 68 / der-064

微分與多變數 / Rank 4/6 Advanced / 表達式 / wronskian / exponential / rank-4

Wronskian $W(e^x, xe^x)$

130 69 / der-065

微分與多變數 / Rank 4/6 Advanced / 表達式 / wronskian / log / rank-4

Wronskian $W(1, \log x)$

131 70 / der-066

微分與多變數 / Rank 4/6 Advanced / 表達式 / wronskian / polynomial / rank-4

Wronskian $W(x^2, x^3)$

132 71 / der-067

微分與多變數 / Rank 5/6 Boss / 表達式 / jacobian-chain / chain-rule / multivariable

If $u = x + y$, $v = x - y$, $F = u^2 + v^2$, find dF/dx

133 72 / der-068

微分與多變數 / Rank 5/6 Boss / 表達式 / jacobian-chain / chain-rule / rank-5

If $u = x^2$, $v = \sin x$, $F = ue^v$, find dF/dx

134 73 / der-069

微分與多變數 / Rank 6/6 Boss+ / 表達式 / jacobian-chain / jacobian / complex

Jacobian determinant of $u = x^2 - y^2$, $v = 2xy$

135 74 / der-070

微分與多變數 / Rank 5/6 Boss / 表達式 / jacobian-chain / jacobian / multivariable

Let $r = u + v$, $s = uv$, $u = x + y$, $v = x - y$. Find $\frac{\partial(r,s)}{\partial(x,y)}$

136 75 / der-071

微分與多變數 / Rank 5/6 Boss / 表達式 / jacobian-chain / jacobian / polar-coordinates

Jacobian $\frac{\partial(x,y)}{\partial(r,t)}$ for $x = r \cos t$, $y = r \sin t$

137 76 / der-072

微分與多變數 / Rank 5/6 Boss / 表達式 / jacobian-chain / jacobian / multivariable

Jacobian $\frac{\partial(u,v)}{\partial(x,y)}$ for $u = xy, v = x/y$

138 77 / der-073

微分與多變數 / Rank 6/6 Boss+ / 表達式 / jacobian-chain / jacobian / complex

Let $u = 2x + y, v = x + 3y, p = u^2 - v^2, q = 2uv$. Find $\frac{\partial(p,q)}{\partial(x,y)}$

139 78 / der-074

微分與多變數 / Rank 5/6 Boss / 數值 / jacobian-chain / jacobian / todai-inspired

If $f(x,y) = (x,xy), g(u,v) = (u+v,u-v)$, find $\det D(g \circ f)$ at $(2,3)$

140 79 / der-075

微分與多變數 / Rank 6/6 Boss+ / 表達式 / complex / real-part / polynomial

Real part of $(x + iy)^3$

141 80 / der-076

微分與多變數 / Rank 6/6 Boss+ / 表達式 / complex / imaginary-part / polynomial

Imaginary part of $(x + iy)^3$

142 81 / der-077

微分與多變數 / Rank 6/6 Boss+ / 數值 / complex / modulus / rank-6

$|1 + i|$

143 82 / der-078

微分與多變數 / Rank 6/6 Boss+ / 數值 / complex / power / rank-6

Real part of $(1 + i)^4$

144 83 / der-079

微分與多變數 / Rank 6/6 Boss+ / 文字 / complex / cauchy-riemann / rank-6

Is $u = x^2 - y^2, v = 2xy$ holomorphic by Cauchy-Riemann?

145 84 / der-080

微分與多變數 / Rank 6/6 Boss+ / 文字 / complex / cauchy-riemann / rank-6

Is $u = x^2 + y^2, v = 0$ holomorphic on an open set?

146 85 / der-081

微分與多變數 / Rank 6/6 Boss+ / 數值 / complex / residue / rank-6

$\left| \operatorname{Res}_{z=i} \frac{1}{z^2 + 1} \right|$

147 86 / der-082

微分與多變數 / Rank 6/6 Boss+ / 數值 / complex / residue / laurent

$$\operatorname{Res}_{z=2} \frac{1}{(z-2)^3}$$

148 87 / der-083

微分與多變數 / Rank 6/6 Boss+ / 數值 / complex / laurent / residue

Coefficient of z^{-1} in e^z/z^3

149 88 / der-084

微分與多變數 / Rank 6/6 Boss+ / 數值 / complex / contour-integral / residue

$$\frac{1}{2\pi i} \int_{|z|=2} \frac{1}{z-1} dz$$

150 89 / der-085

微分與多變數 / Rank 6/6 Boss+ / 數值 / complex / complex-derivative / rank-6

Real part of $f'(1+i)$ for $f(z) = z^4$

151 90 / der-086

微分與多變數 / Rank 6/6 Boss+ / 表達式 / complex / jacobian / cauchy-riemann

Jacobian determinant of the complex map $z \mapsto z^2$ at $z = x + iy$

152 91 / der-087

微分與多變數 / Rank 6/6 Boss+ / 數值 / complex / harmonic / laplacian

$$\Delta \operatorname{Re}(z^3) \quad (z = x + iy)$$

153 92 / der-088

微分與多變數 / Rank 6/6 Boss+ / 文字 / complex / harmonic / laplacian

$$\text{Is } u = x^3 - 3xy^2 \text{ harmonic?}$$

154 93 / gap-der-param-001

微分與多變數 / Rank 2/6 Basic / 表達式 / parametric / parametric-differentiation / rank-2

$$x = t^2, y = t^3. \frac{dy}{dx} = ?$$

155 94 / gap-der-param-002

微分與多變數 / Rank 3/6 Standard / 表達式 / parametric / parametric-differentiation / trig

$$x = \cos t, y = \sin t. \frac{dy}{dx} = ?$$

156 95 / gap-der-param-003

微分與多變數 / Rank 2/6 Basic / 數值 / parametric / second-derivative / rank-2

$$x = t, y = t^2. \frac{d^2y}{dx^2} = ?$$

157 96 / gap-der-param-004

微分與多變數 / Rank 2/6 Basic / 數值 / parametric / parametric-differentiation / rank-2

$$x = t^2 + 1, y = t^3. \left. \frac{dy}{dx} \right|_{t=2}$$

158 97 / gap-der-polar-001

微分與多變數 / Rank 4/6 Advanced / 數值 / polar-curve / parametric / slope

$$r = 1 + \cos \theta. \quad \left. \frac{dy}{dx} \right|_{\theta=\pi/2}$$

159 98 / gap-der-param-005

微分與多變數 / Rank 2/6 Basic / 數值 / parametric / speed / rank-2

$$x = 3t, y = 4t. \text{ Speed } \sqrt{(dx/dt)^2 + (dy/dt)^2}$$

160 99 / gap-der-app-001

微分與多變數 / Rank 3/6 Standard / 數值 / related-rates / volume / rank-3

$$\text{A sphere has } r = 3, dr/dt = 2. \frac{dV}{dt} = ?$$

161 100 / gap-der-app-002

微分與多變數 / Rank 2/6 Basic / 數值 / related-rates / area / rank-2

$$\text{A circle has } r = 5, dr/dt = 0.2. \frac{dA}{dt} = ?$$

162 101 / gap-der-app-003

微分與多變數 / Rank 3/6 Standard / 數值 / related-rates / implicit-differentiation / rank-3

$$\text{Ladder length 10, } x = 6, dx/dt = 2. dy/dt = ?$$

163 102 / gap-der-app-004

微分與多變數 / Rank 2/6 Basic / 數值 / tangent-normal / derivative / rank-2

$$\text{Normal slope to } y = x^2 \text{ at } x = 2$$

164 103 / gap-der-app-005

微分與多變數 / Rank 2/6 Basic / 數值 / linear-approximation / application / rank-2

Linear approximation of $\sqrt{4.04}$

165 104 / gap-der-app-006

微分與多變數 / Rank 2/6 Basic / 數值 / newton-method / application / rank-2

One Newton step for $f(x) = x^2 - 2$ from $x_0 = 1$

166 105 / gap-der-app-007

微分與多變數 / Rank 3/6 Standard / 數值 / curvature / application / rank-3

Curvature of $y = x^2$ at $x = 0$

167 106 / gap-der-ode-001

微分與多變數 / Rank 2/6 Basic / 表達式 / ode-intro / separable / first-order

$y' + y = 0, y(0) = 2. y = ?$

168 107 / gap-der-ode-002

微分與多變數 / Rank 2/6 Basic / 表達式 / ode-intro / separable / exponential-growth

$y' = 2y, y(0) = 3. y = ?$

169 108 / gap-der-ode-003

微分與多變數 / Rank 3/6 Standard / 表達式 / ode-intro / integrating-factor / first-order

$y' + y = e^x, y(0) = 0. y = ?$

170 109 / gap-der-ode-004

微分與多變數 / Rank 3/6 Standard / 表達式 / ode-intro / second-order / trig

$$y'' + y = 0, y(0) = 0, y'(0) = 1. y = ?$$

171 110 / gap-der-ode-005

微分與多變數 / Rank 4/6 Advanced / 表達式 / ode-intro / logistic / separable

$$y' = y(1 - y), y(0) = 1/2. y = ?$$

172 111 / gap-tech-007

微分與多變數 / Rank 2/6 Basic / 文字 / technique-recognition / parametric / rank-2

$$x = t^2, y = t^3. \text{ Which differentiation technique?}$$

173 112 / mob-tech-012

微分與多變數 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

Optimize $f(x, y)$ subject to $g(x, y) = 0$ Use which technique?

174 113 / mob-tech-014

微分與多變數 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$\nabla \cdot \mathbf{F}$ and $\nabla \times \mathbf{F}$ Use which technique?

175 114 / mob-tech-018

微分與多變數 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0 \text{ Use which technique?}$$

176 115 / mob-tech-020

微分與多變數 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$y = x^x$ Use which technique?

177 116 / mob-trap-001

微分與多變數 / Rank 2/6 Basic / 文字 / trap-drill / rank-2 / beginner-friendly

$\frac{d}{dx} \sin(3x)$ Main trap?

178 117 / mob-trap-007

微分與多變數 / Rank 2/6 Basic / 文字 / trap-drill / rank-2 / beginner-friendly

$\frac{\partial}{\partial x} y^2$ Main trap?

179 118 / mob-trap-011

微分與多變數 / Rank 2/6 Basic / 文字 / trap-drill / rank-2 / beginner-friendly

$\nabla \times (\nabla f)$ Main trap?

180 119 / mob-lm-001

微分與多變數 / Rank 4/6 Advanced / 數值 / lagrange-multiplier / optimization / rank-4

$\max xy$ subject to $x + y = 10$ ($x, y > 0$)

181 120 / mob-lm-002

微分與多變數 / Rank 4/6 Advanced / 數值 / lagrange-multiplier / optimization / rank-4

$\min x^2 + y^2$ subject to $x + y = 6$

182 121 / mob-lm-003

微分與多變數 / Rank 4/6 Advanced / 數值 / lagrange-multiplier / optimization / rank-4

$\max xyz$ subject to $x + y + z = 12$ ($x, y, z > 0$)

183 122 / mob-lm-004

微分與多變數 / Rank 4/6 Advanced / 數值 / lagrange-multiplier / optimization / rank-4

$\min x^2 + y^2$ subject to $xy = 8$ ($x, y > 0$)

184 123 / mob-lm-005

微分與多變數 / Rank 4/6 Advanced / 數值 / lagrange-multiplier / optimization / rank-4

$\max x^2y$ subject to $x + 2y = 6$ ($x, y > 0$)

185 124 / mob-lm-006

微分與多變數 / Rank 4/6 Advanced / 數值 / lagrange-multiplier / optimization / rank-4

$\min x^2 + y^2 + z^2$ subject to $x + y + z = 3$

186 125 / mob-lm-007

微分與多變數 / Rank 4/6 Advanced / 數值 / lagrange-multiplier / optimization / rank-4

$\max A$ rectangle with perimeter 20

187 126 / mob-lm-008

微分與多變數 / Rank 4/6 Advanced / 數值 / lagrange-multiplier / optimization / rank-4

$\min x^2 + y^2 + z^2$ subject to $x + y + z = 1$

188 127 / mob-lm-009

微分與多變數 / Rank 4/6 Advanced / 數值 / lagrange-multiplier / optimization / rank-4

$\max xy \text{ subject to } x^2 + y^2 = 50$

189 128 / mob-lm-010

微分與多變數 / Rank 4/6 Advanced / 數值 / lagrange-multiplier / optimization / rank-4

$\max(x + y) \text{ subject to } x^2 + y^2 = 2$

190 129 / mob-bessel-001

微分與多變數 / Rank 6/6 Boss+ / 數值 / bessel / special-functions / rank-6

$J_0(0)$

191 130 / mob-bessel-002

微分與多變數 / Rank 6/6 Boss+ / 數值 / bessel / special-functions / rank-6

$J_1(0)$

192 131 / mob-bessel-003

微分與多變數 / Rank 6/6 Boss+ / 數值 / bessel / special-functions / rank-6

coefficient of x^2 in $J_0(x) = 1 - x^2/4 + \cdots$

coefficient of x in $J_1(x) = x/2 + \cdots$

$J_0'(x)$

$x^2y'' + xy' + (x^2 - \nu^2)y = 0$

Singularity of Bessel equation at $x = 0$

$J_{-2}(x)$ equals

$J_{-1}(x)$ equals

Bessel functions most often appear after which ODE type?

$\nabla \cdot \langle x, y, z \rangle$

z-component of $\nabla \times \langle y, -x, 0 \rangle$

$\Delta(x^2 + y^2 + z^2)$

$D_{(1,1)/\sqrt{2}}(x^2 + y^2)$ at $(1, 1)$

204 143 / mob-nabla-005
微分與多變數 / Rank 4/6 Advanced / 數值 / nabla / vector-calculus / rank-4

$$\nabla \times (\nabla f)$$

205 144 / mob-nabla-006
微分與多變數 / Rank 4/6 Advanced / 數值 / nabla / vector-calculus / rank-4

$$\nabla \cdot (\nabla \times \mathbf{F})$$

206 145 / mob-nabla-007
微分與多變數 / Rank 4/6 Advanced / 數值 / nabla / vector-calculus / rank-4

$$\int_C \nabla(x^2 + y^2) \cdot d\mathbf{r} \text{ from } (0,0) \text{ to } (1,1)$$

207 146 / mob-nabla-008
微分與多變數 / Rank 4/6 Advanced / 文字 / nabla / vector-calculus / rank-4

$$\nabla \cdot \nabla f \text{ is called}$$

208 147 / mob-nabla-009
微分與多變數 / Rank 4/6 Advanced / 文字 / nabla / vector-calculus / rank-4

$$\nabla \times \mathbf{F} = 0 \text{ usually suggests}$$

209 148 / mob-nabla-010
微分與多變數 / Rank 4/6 Advanced / 文字 / nabla / vector-calculus / rank-4

Green theorem circulation uses which scalar derivative?

3 積分 Integrals

共 148 題。題目依 BuzzCalculus 題庫順序排列，適合列印後作為練習本使用。

210 01 / int-001

積分 / Rank 1/6 Warm-up / 不定積分 / rank-1 / beginner-friendly

$$\int 6x^2 \, dx$$

211 02 / int-002

積分 / Rank 1/6 Warm-up / 不定積分 / rank-1 / beginner-friendly

$$\int \cos x \, dx$$

212 03 / int-003

積分 / Rank 1/6 Warm-up / 不定積分 / rank-1 / beginner-friendly

$$\int \frac{1}{x} \, dx$$

213 04 / int-004

積分 / Rank 1/6 Warm-up / 不定積分 / rank-1 / beginner-friendly

$$\int e^{3x} \, dx$$

214 05 / int-005

積分 / Rank 2/6 Basic / 不定積分 / rank-2 / beginner-friendly

$$\int x e^{x^2} \, dx$$

215 06 / int-006

積分 / Rank 2/6 Basic / 不定積分 / rank-2 / beginner-friendly

$$\int \frac{2x}{1+x^2} dx$$

216 07 / int-007

積分 / Rank 2/6 Basic / 不定積分 / rank-2 / beginner-friendly

$$\int x \cos(x^2) dx$$

217 08 / int-008

積分 / Rank 2/6 Basic / 數值 / rank-2 / beginner-friendly

$$\int_0^1 3x^2 dx$$

218 09 / int-009

積分 / Rank 3/6 Standard / 不定積分 / rank-3

$$\int x \ln x dx$$

219 10 / int-010

積分 / Rank 3/6 Standard / 不定積分 / rank-3

$$\int \sin^3 x dx$$

220 11 / int-011

積分 / Rank 3/6 Standard / 不定積分 / rank-3

$$\int \frac{1}{x^2 + 4} dx$$

221 12 / int-012

積分 / Rank 2/6 Basic / 數值 / rank-2 / beginner-friendly

$$\int_0^{\pi/2} \sin^2 x \, dx$$

222 13 / int-013

積分 / Rank 3/6 Standard / 不定積分 / rank-3

$$\int (\ln x)^2 \, dx$$

223 14 / int-014

積分 / Rank 3/6 Standard / 不定積分 / rank-3

$$\int \frac{x^3}{x^2 + 1} \, dx$$

224 15 / int-015

積分 / Rank 4/6 Advanced / 不定積分 / rank-4

$$\int e^x \sin x \, dx$$

225 16 / int-016

積分 / Rank 3/6 Standard / 數值 / rank-3

$$\int_0^1 \ln(1+x) \, dx$$

226 17 / int-017

積分 / Rank 3/6 Standard / 數值 / rank-3

$$\int_0^{\pi/2} x \sin x \, dx$$

227 18 / int-018

積分 / Rank 3/6 Standard / 不定積分 / rank-3

$$\int \frac{\sin x}{1 + \cos x} \, dx$$

228 19 / int-019

積分 / Rank 4/6 Advanced / 不定積分 / rank-4

$$\int \frac{1}{1 + \sqrt{x}} \, dx$$

229 20 / int-020

積分 / Rank 3/6 Standard / 數值 / rank-3

$$\int_0^1 \frac{x^2}{1+x^3} \, dx$$

230 21 / td-int-001

積分 / Rank 6/6 Boss+ / 數值 / rank-6 / boss-rank / boss-plus

$$\int_0^\infty \frac{\cos(3x)}{1+x^2} dx$$

231 22 / td-int-002

積分 / Rank 6/6 Boss+ / 數值 / rank-6 / boss-rank / boss-plus

$$\int_0^\pi \log(5+3\cos x) dx$$

232 23 / td-int-003

積分 / Rank 4/6 Advanced / 數值 / rank-4

$$\int_0^\infty \frac{e^{-5x}-e^{-2x}}{x} dx$$

233 24 / td-int-004

積分 / Rank 4/6 Advanced / 數值 / rank-4

$$\int_0^\infty \frac{\sin(2x)\sin(5x)}{x} dx$$

234 25 / td-int-005

積分 / Rank 6/6 Boss+ / 數值 / rank-6 / boss-rank / boss-plus

$$\text{Area of } 3x^2+2xy+2y^2\leq 1$$

235 26 / td-int-006

積分 / Rank 6/6 Boss+ / 數值 / rank-6 / boss-rank / boss-plus

$$\int_{-\infty}^{\infty} x^4 e^{-3x^2} dx$$

236 27 / int-021

積分 / Rank 4/6 Advanced / 數值 / multivariable / double-integral / rank-4

$$\int_0^1 \int_0^1 (x+y) dy dx$$

237 28 / int-022

積分 / Rank 4/6 Advanced / 數值 / multivariable / area / double-integral

$$\iint_D 1 dA, \quad D = \{x \geq 0, y \geq 0, x+y \leq 2\}$$

238 29 / int-023

積分 / Rank 4/6 Advanced / 數值 / multivariable / double-integral / rank-4

$$\int_0^1 \int_0^x xy dy dx$$

239 30 / int-024

積分 / Rank 4/6 Advanced / 數值 / multivariable / double-integral / triangle

$$\int_0^1 \int_0^{1-x} (x+y) dy dx$$

240 31 / int-025

積分 / Rank 4/6 Advanced / 數值 / multivariable / polar-coordinates / double-integral

$$\iint_{x^2+y^2\leq 1} (x^2+y^2) \, dA$$

241 32 / int-026

積分 / Rank 4/6 Advanced / 數值 / multivariable / double-integral / separable

$$\int_0^2 \int_0^3 xy \, dy \, dx$$

242 33 / int-027

積分 / Rank 2/6 Basic / 不定積分 / substitution / log / rank-2

$$\int \frac{x}{1+x^2} \, dx$$

243 34 / int-028

積分 / Rank 3/6 Standard / 不定積分 / integration-by-parts / log / rank-3

$$\int \log x \, dx$$

244 35 / int-029

積分 / Rank 2/6 Basic / 不定積分 / inverse-trig / standard-integral / rank-2

$$\int \frac{1}{x^2+9} \, dx$$

245 36 / int-030

積分 / Rank 2/6 Basic / 數值 / substitution / definite-integral / rank-2

$$\int_0^{\pi/2} \sin^3 x \cos x \, dx$$

246 37 / int-031

積分 / Rank 4/6 Advanced / 數值 / multivariable / double-integral / change-order

$$\int_0^1 \int_y^1 x^2 \, dx \, dy$$

247 38 / int-032

積分 / Rank 4/6 Advanced / 數值 / multivariable / polar-coordinates / area

$$\iint_{1 \leq x^2 + y^2 \leq 4} 1 \, dA$$

248 39 / int-033

積分 / Rank 4/6 Advanced / 數值 / multivariable / polar-coordinates / symmetry

$$\iint_{x^2 + y^2 \leq 4} x^2 \, dA$$

249 40 / int-034

積分 / Rank 4/6 Advanced / 數值 / multivariable / area / rank-4

$$\int_0^1 \int_0^{\sqrt{1-x^2}} 1 \, dy \, dx$$

250 41 / int-035

積分 / Rank 4/6 Advanced / 數值 / multivariable / polar-coordinates / double-integral

$$\int_0^{2\pi} \int_0^1 r^2 \cdot r \, dr \, d\theta$$

251 42 / int-036

積分 / Rank 4/6 Advanced / 數值 / multivariable / double-integral / triangle

$$\int_0^1 \int_0^{1-y} x \, dx \, dy$$

252 43 / int-037

積分 / Rank 3/6 Standard / 不定積分 / substitution / rank-3

$$\int x e^{x^2} \, dx$$

253 44 / int-038

積分 / Rank 3/6 Standard / 不定積分 / integration-by-parts / rank-3

$$\int x \cos x \, dx$$

254 45 / int-039

積分 / Rank 3/6 Standard / 數值 / inverse-trig / definite-integral / rank-3

$$\int_0^1 \frac{1}{1+x^2} \, dx$$

255 46 / int-040

積分 / Rank 5/6 Boss / 數值 / improper-integral / gamma / rank-5

$$\int_0^\infty x e^{-x} \, dx$$

256 47 / int-041

積分 / Rank 3/6 Standard / 不定積分 / substitution / u-sub / log

$$\int \frac{x^3}{1+x^4} \, dx$$

257 48 / int-042

積分 / Rank 3/6 Standard / 不定積分 / substitution / u-sub / radical

$$\int x^2 \sqrt{1+x^3} \, dx$$

258 49 / int-043

積分 / Rank 4/6 Advanced / 不定積分 / substitution / u-sub / radical

$$\int \frac{x}{\sqrt{1+x^2} \left(1+\sqrt{1+x^2}\right)} \, dx$$

259 50 / int-044

積分 / Rank 3/6 Standard / 不定積分 / substitution / u-sub / exponential

$$\int \frac{x e^{x^2}}{1+e^{x^2}} \, dx$$

260 51 / int-045

積分 / Rank 3/6 Standard / 不定積分 / substitution / u-sub / log

$$\int \frac{(\log x)^2}{x} dx$$

261 52 / int-046

積分 / Rank 3/6 Advanced / 數值 / substitution / u-sub / definite-integral

$$\int_0^1 \frac{2x}{(1+x^2)^3} dx$$

262 53 / int-047

積分 / Rank 3/6 Standard / 數值 / trig-substitution / definite-integral / area

$$\int_0^2 \sqrt{4-x^2} dx$$

263 54 / int-048

積分 / Rank 4/6 Advanced / 數值 / trig-substitution / definite-integral / rank-4

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}} dx$$

264 55 / int-049

積分 / Rank 4/6 Advanced / 數值 / trig-substitution / definite-integral / rank-4

$$\int_0^2 \frac{x^2}{\sqrt{4-x^2}} dx$$

265 56 / int-050

積分 / Rank 4/6 Advanced / 数値 / trig-substitution / inverse-trig / definite-integral

$$\int_1^2 \frac{1}{x\sqrt{x^2-1}} \, dx$$

266 57 / int-051

積分 / Rank 4/6 Advanced / 不定積分 / trig-substitution / radical / rank-4

$$\int \frac{1}{(x^2+4)^{3/2}} \, dx$$

267 58 / int-052

積分 / Rank 3/6 Standard / 不定積分 / integration-by-parts / ibp / exponential

$$\int x^2 e^x \, dx$$

268 59 / int-053

積分 / Rank 4/6 Advanced / 不定積分 / integration-by-parts / ibp / trig

$$\int x^2 \sin x \, dx$$

269 60 / int-054

積分 / Rank 4/6 Advanced / 不定積分 / integration-by-parts / ibp / exponential

$$\int e^x \sin x \, dx$$

270 61 / int-055

積分 / Rank 4/6 Advanced / 不定積分 / integration-by-parts / ibp / log

$$\int (\log x)^2 \, dx$$

271 62 / int-056

積分 / Rank 4/6 Advanced / 數值 / integration-by-parts / ibp / inverse-trig

$$\int_0^1 x \arctan x \, dx$$

272 63 / int-057

積分 / Rank 5/6 Boss / 表達式 / ode-style / convolution / parameter-integral

Find $F(x) = \int_0^x e^{x-t} \sin t \, dt$

273 64 / int-058

積分 / Rank 5/6 Boss / 表達式 / ode-style / convolution / parameter-integral

Find $F(x) = \int_0^x e^{2(x-t)} t \, dt$

274 65 / int-059

積分 / Rank 5/6 Boss / 表達式 / ode-style / convolution / parameter-integral

Find $F(x) = \int_0^x e^{-(x-t)} t^2 \, dt$

275 66 / int-060

積分 / Rank 5/6 Boss / 表達式 / ode-style / convolution / trig

$$\text{Find } F(x) = \int_0^x \sin(x-t) \cos t \, dt$$

276 67 / int-061

積分 / Rank 5/6 Boss / 數值 / kings-property / symmetry / definite-integral

$$\int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} \, dx$$

277 68 / int-062

積分 / Rank 5/6 Boss / 數值 / kings-property / symmetry / definite-integral

$$\int_0^1 \frac{e^x}{e^x + e^{1-x}} \, dx$$

278 69 / int-063

積分 / Rank 5/6 Boss / 數值 / kings-property / symmetry / definite-integral

$$\int_0^{\pi/2} \frac{1 + \sin x}{2 + \sin x + \cos x} \, dx$$

279 70 / int-064

積分 / Rank 5/6 Boss / 數值 / kings-property / symmetry / definite-integral

$$\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$$

280 71 / int-065

積分 / Rank 5/6 Boss / 數值 / frullani / improper-integral / log

$$\int_0^\infty \frac{e^{-3x} - e^{-7x}}{x} dx$$

281 72 / int-066

積分 / Rank 5/6 Boss / 數值 / frullani / improper-integral / log

$$\int_0^\infty \frac{e^{-x} - e^{-4x}}{x} dx$$

282 73 / int-067

積分 / Rank 5/6 Boss / 數值 / frullani / improper-integral / log

$$\int_0^\infty \frac{e^{-2x} - e^{-8x}}{x} dx$$

283 74 / int-068

積分 / Rank 5/6 Boss / 數值 / frullani / cosine-integral / improper-integral

$$\int_0^\infty \frac{\cos(2x) - \cos(5x)}{x} dx$$

284 75 / int-069

積分 / Rank 5/6 Boss / 數值 / frullani / cosine-integral / improper-integral

$$\int_0^\infty \frac{\cos x - \cos(4x)}{x} dx$$

285 76 / int-070

積分 / Rank 5/6 Boss / 數值 / frullani / cosine-integral / improper-integral

$$\int_0^\infty \frac{\cos(3x) - \cos(9x)}{x} dx$$

286 77 / int-071

積分 / Rank 5/6 Boss / 數值 / frullani / trig / improper-integral

$$\int_0^\infty \frac{\sin(3x) \sin(8x)}{x} dx$$

287 78 / int-072

積分 / Rank 5/6 Boss / 數值 / frullani / trig / improper-integral

$$\int_0^\infty \frac{\sin(2x) \sin(6x)}{x} dx$$

288 79 / int-073

積分 / Rank 5/6 Boss / 數值 / frullani / trig / improper-integral

$$\int_0^\infty \frac{\sin x \sin(4x)}{x} dx$$

289 80 / int-074

積分 / Rank 5/6 Boss / 數值 / frullani / inverse-trig / improper-integral

$$\int_0^\infty \frac{\arctan(5x) - \arctan(2x)}{x} dx$$

290 81 / int-075

積分 / Rank 5/6 Boss / 數值 / frullani / rational / improper-integral

$$\int_0^\infty \frac{1}{x} \left(\frac{1}{1+2x} - \frac{1}{1+5x} \right) dx$$

291 82 / int-076

積分 / Rank 5/6 Boss / 數值 / frullani / rational / improper-integral

$$\int_0^\infty \frac{1}{x} \left(\frac{1}{1+x} - \frac{1}{1+6x} \right) dx$$

292 83 / gap-int-pf-001

積分 / Rank 3/6 Standard / 不定積分 / partial-fraction / rational-integral / rank-3

$$\int \frac{1}{x^2 - 1} dx$$

293 84 / gap-int-pf-002

積分 / Rank 3/6 Standard / 不定積分 / partial-fraction / rational-integral / cancellation

$$\int \frac{x + 1}{x^2 + x} dx$$

294 85 / gap-int-pf-003

積分 / Rank 3/6 Standard / 不定積分 / partial-fraction / rational-integral / rank-3

$$\int \frac{1}{x(x + 2)} dx$$

295 86 / gap-int-pf-004

積分 / Rank 3/6 Standard / 不定積分 / partial-fraction / rational-integral / log

$$\int \frac{2x + 3}{x^2 + 3x + 2} dx$$

296 87 / gap-int-pf-005

積分 / Rank 3/6 Standard / 不定積分 / partial-fraction / repeated-factor / rank-3

$$\int \frac{1}{(x+1)^2} dx$$

297 88 / gap-int-pf-006

積分 / Rank 4/6 Advanced / 不定積分 / partial-fraction / rational-integral / rank-4

$$\int \frac{3x+5}{x^2+x-2} dx$$

298 89 / gap-int-pf-007

積分 / Rank 4/6 Advanced / 不定積分 / partial-fraction / rational-integral / rank-4

$$\int \frac{x+4}{x^2+5x+6} dx$$

299 90 / gap-int-polar-001

積分 / Rank 2/6 Basic / 數值 / polar-curve / polar-area / rank-2

Area enclosed by $r = 2$ from $0 \leq \theta \leq \pi/2$

300 91 / gap-int-polar-002

積分 / Rank 3/6 Standard / 數值 / polar-curve / polar-area / trig

Area enclosed by $r = \sin \theta$ for $0 \leq \theta \leq \pi$

301 92 / gap-int-triple-001

積分 / Rank 6/6 Boss+ / 數值 / triple-integral / iterated-integral / rank-6

$$\int_0^1 \int_0^1 \int_0^1 xyz \, dz \, dy \, dx$$

302 93 / gap-int-triple-002

積分 / Rank 6/6 Boss+ / 數值 / triple-integral / iterated-integral / rank-6

$$\int_0^1 \int_0^1 \int_0^1 (x + y + z) \, dz \, dy \, dx$$

303 94 / gap-int-triple-003

積分 / Rank 6/6 Boss+ / 數值 / triple-integral / volume / simplex

$$\iiint_{x+y+z \leq 1, \, x,y,z \geq 0} 1 \, dV$$

304 95 / gap-int-triple-004

積分 / Rank 6/6 Boss+ / 數值 / triple-integral / iterated-integral / rank-6

$$\int_0^1 \int_0^x \int_0^y 1 \, dz \, dy \, dx$$

305 96 / gap-int-triple-005
積分 / Rank 6/6 Boss+ / 數值 / triple-integral / cylindrical-coordinates / volume

Volume of the cylinder $x^2 + y^2 \leq 4, 0 \leq z \leq 3$

306 97 / gap-int-triple-006
積分 / Rank 6/6 Boss+ / 數值 / triple-integral / spherical-coordinates / volume

Volume of the ball $x^2 + y^2 + z^2 \leq 4$

307 98 / gap-int-cov-001
積分 / Rank 6/6 Boss+ / 數值 / change-of-variables / jacobian / rank-6

$u = x + y, v = x - y. \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = ?$

308 99 / gap-int-cov-002
積分 / Rank 6/6 Boss+ / 數值 / change-of-variables / jacobian / area

Area of $0 \leq x + y \leq 1, 0 \leq x - y \leq 2$

309 100 / gap-tech-003

積分 / Rank 2/6 Basic / 文字 / technique-recognition / substitution / u-sub

Technique for $\int x e^{x^2} dx$

310 101 / gap-tech-004

積分 / Rank 3/6 Standard / 文字 / technique-recognition / integration-by-parts / ibp

Technique for $\int x \log x dx$

311 102 / gap-tech-006

積分 / Rank 3/6 Standard / 文字 / technique-recognition / partial-fraction / rank-3

Technique for $\int \frac{1}{x^2 - 1} dx$

312 103 / mob-tech-003

積分 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$\int x e^{x^2} dx$ Use which technique?

313 104 / mob-tech-004

積分 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$\int x \log x dx$ Use which technique?

314 105 / mob-tech-005

積分 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$\int \frac{dx}{x^2 - 1}$ Use which technique?

315 106 / mob-tech-006

積分 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$\int \frac{dx}{\sqrt{a^2 - x^2}}$ Use which technique?

316 107 / mob-tech-010

積分 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$\int_0^1 f(x) dx$ with $f(x) + f(1 - x)$ Use which technique?

317 108 / mob-tech-011

積分 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} dx$ Use which technique?

318 109 / mob-tech-013

積分 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$\int_C \nabla \phi \cdot d\mathbf{r}$ Use which technique?

319 110 / mob-tech-015

積分 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$$\int_0^1 x^{a-1}(1-x)^{b-1} dx$$
 Use which technique?

320 111 / mob-tech-016

積分 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$$\int_0^\infty x^{s-1}e^{-x} dx$$
 Use which technique?

321 112 / mob-tech-017

積分 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$$\int_0^{\pi/2} \sin^n x dx$$
 Use which technique?

322 113 / mob-trap-002

積分 / Rank 2/6 Basic / 文字 / trap-drill / rank-2 / beginner-friendly

$$\int \cos(3x) dx$$
 Main trap?

323 114 / mob-trap-003

積分 / Rank 2/6 Basic / 文字 / trap-drill / rank-2 / beginner-friendly

$$\int \frac{1}{x^2+1} dx$$
 Main trap?

324 115 / mob-trap-006

積分 / Rank 2/6 Basic / 文字 / trap-drill / rank-2 / beginner-friendly

$$\int_0^\infty e^{-ax} \, dx \quad \text{Main trap?}$$

325 116 / mob-trap-008

積分 / Rank 2/6 Basic / 文字 / trap-drill / rank-2 / beginner-friendly

$$\iint_R f(x,y) \, dA \quad \text{under change of variables} \quad \text{Main trap?}$$

326 117 / mob-trap-010

積分 / Rank 2/6 Basic / 文字 / trap-drill / rank-2 / beginner-friendly

$$\int \sec x \tan x \, dx \quad \text{Main trap?}$$

327 118 / mob-trap-012

積分 / Rank 2/6 Basic / 文字 / trap-drill / rank-2 / beginner-friendly

$$B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \quad \text{Main trap?}$$

328 119 / mob-beta-001

積分 / Rank 5/6 Boss / 數值 / beta-function / special-functions / rank-5

$$B(1,1)$$

$B(2,1)$

$B(2,2)$

$B(3,2)$

$B(1/2,1/2)$

333 124 / mob-beta-006
積分 / Rank 5/6 Boss / 數值 / beta-function / special-functions / rank-5

$$B(1/2,1)$$

334 125 / mob-beta-007
積分 / Rank 5/6 Boss / 數值 / beta-function / special-functions / rank-5

$$B(3/2,1/2)$$

335 126 / mob-beta-008
積分 / Rank 5/6 Boss / 數值 / beta-function / special-functions / rank-5

$$\int_0^1 \sqrt{x(1-x)}\,dx$$

336 127 / mob-beta-009
積分 / Rank 5/6 Boss / 數值 / beta-function / special-functions / rank-5

$$B(3,3)$$

$B(4,1)$

$\Gamma(1)$

$\Gamma(1/2)$

$\Gamma(3/2)$

341 132 / mob-gamma-004

積分 / Rank 5/6 Boss / 數值 / gamma-function / special-functions / rank-5

$$\Gamma(5/2)$$

342 133 / mob-gamma-005

積分 / Rank 5/6 Boss / 數值 / gamma-function / special-functions / rank-5

$$\Gamma(6)$$

343 134 / mob-gamma-006

積分 / Rank 5/6 Boss / 數值 / gamma-function / special-functions / rank-5

$$\int_0^\infty x^2 e^{-x} dx$$

344 135 / mob-gamma-007

積分 / Rank 5/6 Boss / 數值 / gamma-function / special-functions / rank-5

$$\int_0^\infty x^{1/2} e^{-x} dx$$

345 136 / mob-gamma-008

積分 / Rank 5/6 Boss / 數值 / gamma-function / special-functions / rank-5

$\Gamma(4)$

346 137 / mob-gamma-009

積分 / Rank 5/6 Boss / 數值 / gamma-function / special-functions / rank-5

$\Gamma(7/2)$

347 138 / mob-gamma-010

積分 / Rank 5/6 Boss / 數值 / gamma-function / special-functions / rank-5

$\Gamma(0+1)$

348 139 / mob-wallis-001

積分 / Rank 5/6 Boss / 數值 / wallis / special-functions / rank-5

$\int_0^{\pi/2} \sin^2 x \, dx$

349 140 / mob-wallis-002

積分 / Rank 5/6 Boss / 數值 / wallis / special-functions / rank-5

$$\int_0^{\pi/2} \sin^3 x \, dx$$

350 141 / mob-wallis-003

積分 / Rank 5/6 Boss / 數值 / wallis / special-functions / rank-5

$$\int_0^{\pi/2} \sin^4 x \, dx$$

351 142 / mob-wallis-004

積分 / Rank 5/6 Boss / 數值 / wallis / special-functions / rank-5

$$\int_0^{\pi/2} \sin^5 x \, dx$$

352 143 / mob-wallis-005

積分 / Rank 5/6 Boss / 數值 / wallis / special-functions / rank-5

$$\int_0^{\pi/2} \cos^2 x \, dx$$

353 144 / mob-wallis-006

積分 / Rank 5/6 Boss / 數值 / wallis / special-functions / rank-5

$$\int_0^{\pi/2} \cos^4 x \, dx$$

354 145 / mob-wallis-007

積分 / Rank 5/6 Boss / 數值 / wallis / special-functions / rank-5

$$\int_0^{\pi/2} \sin^2 x \cos^2 x \, dx$$

355 146 / mob-wallis-008

積分 / Rank 5/6 Boss / 數值 / wallis / special-functions / rank-5

$$\int_0^{\pi/2} \sin x \, dx$$

356 147 / mob-wallis-009

積分 / Rank 5/6 Boss / 數值 / wallis / special-functions / rank-5

$$\int_0^{\pi/2} \cos x \, dx$$

$$\int_0^{\pi/2} \sin x \cos x \, dx$$

4 級數與收斂 Series

共 69 題。題目依 BuzzCalculus 題庫順序排列，適合列印後作為練習本使用。

358 01 / ser-001
級數與收斂 / Rank 1/6 Warm-up / 文字 / rank-1 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

359 02 / ser-002
級數與收斂 / Rank 1/6 Warm-up / 文字 / rank-1 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

360 03 / ser-003
級數與收斂 / Rank 1/6 Warm-up / 數值 / rank-1 / beginner-friendly

$$\sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n$$

361 04 / ser-004
級數與收斂 / Rank 1/6 Warm-up / 文字 / rank-1 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

362 05 / ser-005

級數與收斂 / Rank 2/6 Basic / 文字 / rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{n}{n^2+1}$$

363 06 / ser-006

級數與收斂 / Rank 2/6 Basic / 文字 / rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}$$

364 07 / ser-007

級數與收斂 / Rank 2/6 Basic / 數值 / rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{3^n}{n!}$$

365 08 / ser-008

級數與收斂 / Rank 2/6 Basic / 數值 / rank-2 / beginner-friendly

Radius of convergence of $\sum_{n=0}^{\infty} n!x^n$

366 09 / ser-009

級數與收斂 / Rank 3/6 Standard / 數值 / rank-3

Radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$

367 10 / ser-010

級數與收斂 / Rank 3/6 Standard / 文字 / rank-3

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$$

368 11 / ser-011

級數與收斂 / Rank 3/6 Standard / 數值 / rank-3

$$\sum_{n=1}^{\infty} \frac{2^n + n}{3^n}$$

369 12 / ser-012

級數與收斂 / Rank 2/6 Basic / 數值 / rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

370 13 / ser-013

級數與收斂 / Rank 2/6 Basic / 文字 / rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} (-1)^n \frac{n}{n+1}$$

371 14 / ser-014

級數與收斂 / Rank 3/6 Standard / 數值 / rank-3

$$\sum_{n=1}^{\infty} \frac{n}{2^n}$$

372 15 / ser-015

級數與收斂 / Rank 3/6 Standard / 數值 / rank-3

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n}$$

373 16 / ser-016

級數與收斂 / Rank 4/6 Advanced / 文字 / rank-4

$$\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}$$

374 17 / ser-017

級數與收斂 / Rank 3/6 Standard / 數值 / rank-3

Radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 x^n}{3^n}$

375 18 / ser-018

級數與收斂 / Rank 2/6 Basic / 文字 / rank-2 / beginner-friendly

$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$

376 19 / ser-019

級數與收斂 / Rank 2/6 Basic / 數值 / rank-2 / beginner-friendly

$\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$

377 20 / ser-020

級數與收斂 / Rank 3/6 Standard / 文字 / rank-3

$\sum_{n=2}^{\infty} \frac{\ln n}{n}$

378 21 / td-ser-001

級數與收斂 / Rank 6/6 Boss+ / 數值 / rank-6 / boss-rank / boss-plus

Coefficient of x^6 in $(1+x)^x$

379 22 / td-ser-002

級數與收斂 / Rank 6/6 Boss+ / 數值 / rank-6 / boss-rank / boss-plus

Coefficient of x^5 in $e^{2 \arcsin x}$

380 23 / td-ser-003

級數與收斂 / Rank 6/6 Boss+ / 數值 / rank-6 / boss-rank / boss-plus

$$\sum_{n=1}^{\infty} \frac{(2n-1)!!}{(2n)!!} \cdot \frac{1}{n}$$

381 24 / td-ser-004

級數與收斂 / Rank 4/6 Advanced / 數值 / rank-4

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n}$$

382 25 / ser-021

級數與收斂 / Rank 2/6 Basic / 數值 / telescoping / rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

383 26 / ser-022

級數與收斂 / Rank 2/6 Basic / 數值 / power-series / geometric-derivative / rank-2

$$\sum_{n=1}^{\infty} \frac{n}{2^n}$$

384 27 / ser-023

級數與收斂 / Rank 2/6 Basic / 文字 / alternating-series / conditional-convergence / rank-2

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

385 28 / ser-024

級數與收斂 / Rank 2/6 Basic / 文字 / integral-test / log / rank-2

$$\sum_{n=2}^{\infty} \frac{1}{n \log n}$$

386 29 / ser-025

級數與收斂 / Rank 3/6 Standard / 文字 / ratio-test / rank-3

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}$$

387 30 / ser-026

級數與收斂 / Rank 2/6 Basic / 數值 / power-series / radius / rank-2

Radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 x^n}{3^n}$

388 31 / ser-027

級數與收斂 / Rank 2/6 Basic / 數值 / taylor / coefficient / rank-2

Coefficient of x^4 in e^{2x}

389 32 / ser-028

級數與收斂 / Rank 1/6 Warm-up / 文字 / p-series / rank-1 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

390 33 / ser-029

級數與收斂 / Rank 2/6 Basic / 數值 / exponential-series / rank-2 / beginner-friendly

$$\sum_{n=1}^{\infty} \frac{3^n}{n!}$$

391 34 / ser-030

級數與收斂 / Rank 2/6 Basic / 數值 / power-series / radius / rank-2

Radius of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n5^n}$

392 35 / ser-031

級數與收斂 / Rank 3/6 Standard / 文字 / p-series / rank-3

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

393 36 / ser-032

級數與收斂 / Rank 3/6 Standard / 文字 / term-test / rank-3

$$\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^n$$

394 37 / ser-033

級數與收斂 / Rank 3/6 Standard / 文字 / alternating-series / conditional-convergence / rank-3

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$$

395 38 / ser-034

級數與收斂 / Rank 3/6 Standard / 數值 / power-series / radius / ratio-test

Radius of convergence of $\sum_{n=1}^{\infty} n!x^n$

396 39 / ser-035

級數與收斂 / Rank 3/6 Standard / 文字 / power-series / radius / ratio-test

Radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n!}$

397 40 / ser-036

級數與收斂 / Rank 3/6 Standard / 文字 / comparison / geometric-series / rank-3

$$\sum_{n=1}^{\infty} \frac{2^n}{3^n + n}$$

398 41 / ser-037

級數與收斂 / Rank 3/6 Standard / 文字 / integral-test / log / rank-3

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^2}$$

399 42 / ser-038

級數與收斂 / Rank 2/6 Basic / 數值 / taylor / coefficient / rank-2

Coefficient of x^3 in $\log(1+x)$

400 43 / ser-039

級數與收斂 / Rank 3/6 Standard / 數值 / power-series / radius / rank-3

Radius of convergence of $\sum_{n=1}^{\infty} \frac{(2x)^n}{n}$

401 44 / ser-040

級數與收斂 / Rank 3/6 Standard / 數值 / power-series / geometric-derivative / rank-3

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n}$$

402 45 / gap-ser-root-001

級數與收斂 / Rank 3/6 Standard / 數值 / root-test / power-series / radius

Radius of convergence of $\sum_{n=1}^{\infty} \frac{3^n}{n^2} x^n$

403 46 / gap-ser-root-002

級數與收斂 / Rank 3/6 Standard / 數值 / root-test / power-series / radius

Radius of convergence of $\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^n x^n$

404 47 / gap-ser-lc-001

級數與收斂 / Rank 3/6 Standard / 文字 / limit-comparison / harmonic / rank-3

$\sum_{n=1}^{\infty} \frac{3n^2 + 1}{n^3 + 2}$ converges or diverges?

405 48 / gap-ser-lc-002

級數與收斂 / Rank 3/6 Standard / 文字 / limit-comparison / p-series / rank-3

$\sum_{n=1}^{\infty} \frac{n+1}{n^3+1}$ converges or diverges?

406 49 / gap-ser-cond-001

級數與收斂 / Rank 3/6 Standard / 文字 / alternating-series / absolute-conditional / rank-3

$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ absolute, conditional, or divergent?

407 50 / gap-ser-end-001

級數與收斂 / Rank 4/6 Advanced / 文字 / endpoint-analysis / power-series / alternating-series

$\sum_{n=1}^{\infty} \frac{x^n}{n}$ at $x = -1$ is absolute, conditional, or divergent?

408 51 / gap-ser-end-002

級數與收斂 / Rank 4/6 Advanced / 文字 / endpoint-analysis / power-series / harmonic

$\sum_{n=1}^{\infty} \frac{x^n}{n}$ at $x = 1$ converges or diverges?

409 52 / gap-ser-taylor-001

級數與收斂 / Rank 3/6 Standard / 數值 / taylor / coefficient / full-expansion

Coefficient of x^5 in the Maclaurin series for $\sin x$

410 53 / gap-ser-taylor-002

級數與收斂 / Rank 3/6 Standard / 表達式 / taylor / full-expansion / rank-3

Degree 4 Maclaurin polynomial for $\cos x$

411 54 / gap-tech-005

級數與收斂 / Rank 3/6 Standard / 文字 / technique-recognition / root-test / rank-3

Technique for $\sum \left(\frac{n}{n+1}\right)^n x^n$

412 55 / mob-tech-007

級數與收斂 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$\sum_{n=1}^{\infty} \frac{n!}{3^n}$ Use which technique?

413 56 / mob-tech-008

級數與收斂 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

$\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ Use which technique?

414 57 / mob-tech-019

級數與收斂 / Rank 2/6 Basic / 文字 / technique-sprint / technique-recognition / rank-2

Find interval of convergence after radius is known Use which technique?

415 58 / mob-trap-004

級數與收斂 / Rank 2/6 Basic / 文字 / trap-drill / rank-2 / beginner-friendly

$\sum (-1)^n/n$ is not absolutely convergent Main trap?

416 59 / mob-trap-005

級數與收斂 / Rank 2/6 Basic / 文字 / trap-drill / rank-2 / beginner-friendly

$\sum x^n/n$ after finding $R = 1$ Main trap?

417 60 / mob-conv-001

級數與收斂 / Rank 3/6 Standard / 文字 / convergence-test / series-test / rank-3

$\sum_{n=1}^{\infty} \frac{1}{n}$

418 61 / mob-conv-002

級數與收斂 / Rank 3/6 Standard / 文字 / convergence-test / series-test / rank-3

$\sum_{n=1}^{\infty} \frac{1}{n^2}$

419 62 / mob-conv-003

級數與收斂 / Rank 3/6 Standard / 文字 / convergence-test / series-test / rank-3

$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

420 63 / mob-conv-004

級數與收斂 / Rank 3/6 Standard / 文字 / convergence-test / series-test / rank-3

$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$

421 64 / mob-conv-005

級數與收斂 / Rank 3/6 Standard / 文字 / convergence-test / series-test / rank-3

$$\sum_{n=1}^{\infty} \frac{n!}{n^n}$$

422 65 / mob-conv-006

級數與收斂 / Rank 3/6 Standard / 文字 / convergence-test / series-test / rank-3

$$\sum_{n=1}^{\infty} \frac{2^n}{n!}$$

423 66 / mob-conv-007

級數與收斂 / Rank 3/6 Standard / 文字 / convergence-test / series-test / rank-3

$$\sum_{n=1}^{\infty} \frac{n}{n^2 + 1}$$

424 67 / mob-conv-008

級數與收斂 / Rank 3/6 Standard / 文字 / convergence-test / series-test / rank-3

$$\sum_{n=2}^{\infty} \frac{1}{n \log n}$$

425 68 / mob-conv-009

級數與收斂 / Rank 3/6 Standard / 文字 / convergence-test / series-test / rank-3

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^2}$$

426 69 / mob-conv-010

級數與收斂 / Rank 3/6 Standard / 文字 / convergence-test / series-test / rank-3

$$\sum_{n=1}^{\infty} \frac{n+1}{3n+2}$$

答案索引

- 001. lim-001 1
- 002. lim-002 $1/2$
- 003. lim-003 4
- 004. lim-004 $3/2$
- 005. lim-005 1
- 006. lim-006 3
- 007. lim-007 2
- 008. lim-008 $1/2$
- 009. lim-009 $-1/6$
- 010. lim-010 2
- 011. lim-011 0
- 012. lim-012 $1/3$
- 013. lim-013 $1/2$
- 014. lim-014 $-1/2$
- 015. lim-015 $1/2$
- 016. lim-016 $-1/8$
- 017. lim-017 $1/3$
- 018. lim-018 $1/3$
- 019. lim-019 $-e/2$
- 020. lim-020 $-1/3$
- 021. td-lim-001 $\exp(1/2)$
- 022. td-lim-002 $\sqrt{e}/3$
- 023. td-lim-003 $11*e/24$
- 024. td-lim-004 $-1/2$
- 025. lim-021 $1/3$
- 026. lim-022 $3/2$
- 027. lim-023 2
- 028. lim-024 $-1/2$
- 029. lim-025 1
- 030. lim-026 0
- 031. lim-027 dne
- 032. lim-028 $1/2$
- 033. lim-029 0
- 034. lim-030 $9/2$
- 035. lim-031 0
- 036. lim-032 dne
- 037. lim-033 1
- 038. lim-034 0

039. lim-035 $5/2$
 040. lim-036 $1/2$
 041. lim-037 2
 042. lim-038 $1/3$
 043. lim-039 dne
 044. lim-040 2
 045. gap-tech-001 rationalize
 046. gap-tech-002 taylor
 047. gap-tech-008 path test
 048. mob-tech-001 rationalize
 049. mob-tech-002 Taylor
 050. mob-tech-009 path test
 051. mob-trap-009 indeterminate
 052. mob-limtrap-001 $-1/6$
 053. mob-limtrap-002 $1/3$
 054. mob-limtrap-003 $-1/24$
 055. mob-limtrap-004 $1/3$
 056. mob-limtrap-005 $1/6$
 057. mob-limtrap-006 $-1/8$
 058. mob-limtrap-007 0
 059. mob-limtrap-008 0
 060. mob-limtrap-009 DNE
 061. mob-limtrap-010 DNE
 062. der-001 $5x^4-6x$
 063. der-002 $\cos(x)-\sin(x)$
 064. der-003 $\exp(x)*\log(x)+\exp(x)/x$
 065. der-004 $1/(2*\sqrt{x})$
 066. der-005 $2x/(1+x^2)$
 067. der-006 $2x*\sin(x)+x^2*\cos(x)$
 068. der-007 $1/(1+x^2)$
 069. der-008 $1/(x+1)^2$
 070. der-009 $x^x*(\log(x)+1)$
 071. der-010 $\exp(-x)*(2*x*\cos(x^2)-\sin(x^2))$
 072. der-011 $\cos(x)/\sin(x)$
 073. der-012 $((x^2+1)/x-2*x*\log(x))/(x^2+1)^2$
 074. der-013 $x^{\sin(x)}*(\cos(x)*\log(x)+\sin(x)/x)$
 075. der-014 $1/(1+x^2)$
 076. der-015 $1/\sqrt{1+x^2}$
 077. der-016 $1/(1+\cos(x))$
 078. der-017 $\exp(x^2)*(2*x*\cos(3*x)-3*\sin(3*x))$

079. der-018 $1/\sqrt{x^2-1}$
 080. der-019 $10*x*(x^2+1)^4$
 081. der-020 $1/(2*\sqrt{x}*(1+x))$
 082. td-der-001 $2*(1-3*x^2)/(1+x^2)^3$
 083. td-der-002 $(35*x^3-15*x)/2$
 084. td-der-003 $\log(1+x)+x/(1+x)$
 085. td-der-004 $-3*\sin(x)/(5+3*\cos(x))$
 086. der-021 $2*x*y+y*\cos(x*y)$
 087. der-022 $x^2*\exp(x*y)+3*y^2$
 088. der-023 $\exp(x*y)*(1+x*y)$
 089. der-024 $22/5$
 090. der-025 $2*x/(x^2+y^2)$
 091. der-026 $2*x*\sin(3*x)+3*x^2*\cos(3*x)$
 092. der-027 $x/(1+x^2)$
 093. der-028 $\exp(x)/(1+\exp(2*x))$
 094. der-029 $\exp(x)*(x+2)$
 095. der-030 $-(2*x+y)/(x+2*y)$
 096. der-031 $2*x*\exp(x^2+y^2)$
 097. der-032 $x/(x^2+y^2)$
 098. der-033 $2*y-\sin(x)$
 099. der-034 $x-y$
 100. der-035 $\exp(t)*(4*t^3+t^4)$
 101. der-036 4
 102. der-037 27
 103. der-038 $(x^2-2*x-1)/(x-1)^2$
 104. der-039 $2*\sin(x)*\cos(x)$
 105. der-040 $2*x^{\log(x)}*\log(x)/x$
 106. der-041 -0.04
 107. der-042 -0.028
 108. der-043 0
 109. der-044 $y*\exp(x*y)*dx+x*\exp(x*y)*dy$
 110. der-045 $-y*dx/(x^2+y^2)+x*dy/(x^2+y^2)$
 111. der-046 4.92
 112. der-047 $-9/2$
 113. der-048 -9
 114. der-049 -1
 115. der-050 5
 116. der-051 3
 117. der-052 144
 118. der-053 -1

119. der-054 128
 120. der-055 local minimum
 121. der-056 saddle point
 122. der-057 0
 123. der-058 -4
 124. der-059 -1
 125. der-060 1
 126. der-061 $\exp(3x)$
 127. der-062 x^2
 128. der-063 2
 129. der-064 $\exp(2x)$
 130. der-065 $1/x$
 131. der-066 x^4
 132. der-067 $4x$
 133. der-068 $\exp(\sin(x)) \cdot (2x + x^2 \cos(x))$
 134. der-069 $4(x^2 + y^2)$
 135. der-070 $-4y$
 136. der-071 r
 137. der-072 $-2x/y$
 138. der-073 $20((2x+y)^2 + (x+3y)^2)$
 139. der-074 -4
 140. der-075 $x^3 - 3xy^2$
 141. der-076 $3x^2y - y^3$
 142. der-077 $\sqrt{2}$
 143. der-078 -4
 144. der-079 holomorphic
 145. der-080 not holomorphic
 146. der-081 $1/2$
 147. der-082 0
 148. der-083 $1/2$
 149. der-084 1
 150. der-085 -8
 151. der-086 $4(x^2 + y^2)$
 152. der-087 0
 153. der-088 harmonic
 154. gap-der-param-001 $3t/2$
 155. gap-der-param-002 $-\cos(t)/\sin(t)$
 156. gap-der-param-003 2
 157. gap-der-param-004 3
 158. gap-der-polar-001 1

159. gap-der-param-005 5
 160. gap-der-app-001 72π
 161. gap-der-app-002 2π
 162. gap-der-app-003 $-3/2$
 163. gap-der-app-004 $-1/4$
 164. gap-der-app-005 2.01
 165. gap-der-app-006 $3/2$
 166. gap-der-app-007 2
 167. gap-der-ode-001 $2\exp(-x)$
 168. gap-der-ode-002 $3\exp(2x)$
 169. gap-der-ode-003 $(\exp(x)-\exp(-x))/2$
 170. gap-der-ode-004 $\sin(x)$
 171. gap-der-ode-005 $1/(1+\exp(-x))$
 172. gap-tech-007 parametric differentiation
 173. mob-tech-012 Lagrange multiplier
 174. mob-tech-014 nabla
 175. mob-tech-018 Bessel equation
 176. mob-tech-020 logarithmic differentiation
 177. mob-trap-001 missing chain factor
 178. mob-trap-007 treat y constant
 179. mob-trap-011 zero
 180. mob-lm-001 25
 181. mob-lm-002 18
 182. mob-lm-003 64
 183. mob-lm-004 16
 184. mob-lm-005 16
 185. mob-lm-006 3
 186. mob-lm-007 25
 187. mob-lm-008 $1/3$
 188. mob-lm-009 25
 189. mob-lm-010 2
 190. mob-bessel-001 1
 191. mob-bessel-002 0
 192. mob-bessel-003 $-1/4$
 193. mob-bessel-004 $1/2$
 194. mob-bessel-005 $-J_1(x)$
 195. mob-bessel-006 Bessel equation
 196. mob-bessel-007 regular singular
 197. mob-bessel-008 $J_2(x)$
 198. mob-bessel-009 $-J_1(x)$

199. mob-bessel-010 second-order ODE
 200. mob-nabla-001 3
 201. mob-nabla-002 -2
 202. mob-nabla-003 6
 203. mob-nabla-004 $2\sqrt{2}$
 204. mob-nabla-005 0
 205. mob-nabla-006 0
 206. mob-nabla-007 2
 207. mob-nabla-008 Laplacian
 208. mob-nabla-009 conservative field
 209. mob-nabla-010 curl
 210. int-001 $2x^3$
 211. int-002 $\sin(x)$
 212. int-003 $\log(\text{abs}(x))$
 213. int-004 $\exp(3x)/3$
 214. int-005 $\exp(x^2)/2$
 215. int-006 $\log(1+x^2)$
 216. int-007 $\sin(x^2)/2$
 217. int-008 1
 218. int-009 $x^2 \log(x)/2 - x^2/4$
 219. int-010 $-\cos(x) + \cos(x)^3/3$
 220. int-011 $\text{atan}(x/2)/2$
 221. int-012 $\pi/4$
 222. int-013 $x(\log(x)^2 - 2\log(x) + 2)$
 223. int-014 $x^2/2 - \log(x^2 + 1)/2$
 224. int-015 $\exp(x)(\sin(x) - \cos(x))/2$
 225. int-016 $2\log(2) - 1$
 226. int-017 1
 227. int-018 $-\log(1 + \cos(x))$
 228. int-019 $2\sqrt{x} - 2\log(1 + \sqrt{x})$
 229. int-020 $\log(2)/3$
 230. td-int-001 $\pi \exp(-3)/2$
 231. td-int-002 $\pi \log(9/2)$
 232. td-int-003 $\log(2/5)$
 233. td-int-004 $\log(7/3)/2$
 234. td-int-005 $\pi/\sqrt{5}$
 235. td-int-006 $\sqrt{\pi}/(12\sqrt{3})$
 236. int-021 1
 237. int-022 2
 238. int-023 $1/8$

239. int-024 $1/3$
 240. int-025 $\pi/2$
 241. int-026 9
 242. int-027 $\log(1+x^2)/2$
 243. int-028 $x \cdot \log(x) - x$
 244. int-029 $\operatorname{atan}(x/3)/3$
 245. int-030 $1/4$
 246. int-031 $1/4$
 247. int-032 3π
 248. int-033 4π
 249. int-034 $\pi/4$
 250. int-035 $\pi/2$
 251. int-036 $1/6$
 252. int-037 $\exp(x^2)/2$
 253. int-038 $x \cdot \sin(x) + \cos(x)$
 254. int-039 $\pi/4$
 255. int-040 1
 256. int-041 $\log(1+x^4)/4$
 257. int-042 $(2/9) \cdot (1+x^3)^{(3/2)}$
 258. int-043 $\log(1+\sqrt{1+x^2})$
 259. int-044 $\log(1+\exp(x^2))/2$
 260. int-045 $\log(x)^3/3$
 261. int-046 $3/8$
 262. int-047 π
 263. int-048 $\pi/4$
 264. int-049 π
 265. int-050 $\pi/3$
 266. int-051 $x/(4 \cdot \sqrt{x^2+4})$
 267. int-052 $\exp(x) \cdot (x^2 - 2x + 2)$
 268. int-053 $-x^2 \cdot \cos(x) + 2x \cdot \sin(x) + 2 \cdot \cos(x)$
 269. int-054 $\exp(x) \cdot (\sin(x) - \cos(x))/2$
 270. int-055 $x \cdot (\log(x)^2 - 2 \cdot \log(x) + 2)$
 271. int-056 $\pi/4 - 1/2$
 272. int-057 $(\exp(x) - \sin(x) - \cos(x))/2$
 273. int-058 $(\exp(2x) - 2x - 1)/4$
 274. int-059 $x^2 - 2x + 2 - 2 \cdot \exp(-x)$
 275. int-060 $x \cdot \sin(x)/2$
 276. int-061 $\pi/4$
 277. int-062 $1/2$
 278. int-063 $\pi/4$

279. int-064 $\pi^2/4$
 280. int-065 $\log(7/3)$
 281. int-066 $\log(4)$
 282. int-067 $\log(4)$
 283. int-068 $\log(5/2)$
 284. int-069 $\log(4)$
 285. int-070 $\log(3)$
 286. int-071 $\log(11/5)/2$
 287. int-072 $\log(2)/2$
 288. int-073 $\log(5/3)/2$
 289. int-074 $\pi \cdot \log(5/2)/2$
 290. int-075 $\log(5/2)$
 291. int-076 $\log(6)$
 292. gap-int-pf-001 $(\log(x-1)-\log(x+1))/2$
 293. gap-int-pf-002 $\log(x)$
 294. gap-int-pf-003 $(\log(x)-\log(x+2))/2$
 295. gap-int-pf-004 $\log(x+1)+\log(x+2)$
 296. gap-int-pf-005 $-1/(x+1)$
 297. gap-int-pf-006 $8 \cdot \log(x-1)/3 + \log(x+2)/3$
 298. gap-int-pf-007 $2 \cdot \log(x+2) - \log(x+3)$
 299. gap-int-polar-001 π
 300. gap-int-polar-002 $\pi/4$
 301. gap-int-triple-001 $1/8$
 302. gap-int-triple-002 $3/2$
 303. gap-int-triple-003 $1/6$
 304. gap-int-triple-004 $1/6$
 305. gap-int-triple-005 $12 \cdot \pi$
 306. gap-int-triple-006 $32 \cdot \pi/3$
 307. gap-int-cov-001 $1/2$
 308. gap-int-cov-002 1
 309. gap-tech-003 substitution
 310. gap-tech-004 integration by parts
 311. gap-tech-006 partial fraction
 312. mob-tech-003 u-substitution
 313. mob-tech-004 integration by parts
 314. mob-tech-005 partial fraction
 315. mob-tech-006 trig substitution
 316. mob-tech-010 King's property
 317. mob-tech-011 Frullani integral
 318. mob-tech-013 gradient theorem

319. mob-tech-015 Beta function
 320. mob-tech-016 Gamma function
 321. mob-tech-017 Wallis reduction
 322. mob-trap-002 divide by inner coefficient
 323. mob-trap-003 arctan not log
 324. mob-trap-006 requires a positive
 325. mob-trap-008 multiply by Jacobian
 326. mob-trap-010 $\sec x$
 327. mob-trap-012 denominator $\Gamma(a+b)$
 328. mob-beta-001 1
 329. mob-beta-002 $1/2$
 330. mob-beta-003 $1/6$
 331. mob-beta-004 $1/12$
 332. mob-beta-005 π
 333. mob-beta-006 2
 334. mob-beta-007 $\pi/2$
 335. mob-beta-008 $\pi/8$
 336. mob-beta-009 $1/30$
 337. mob-beta-010 $1/4$
 338. mob-gamma-001 1
 339. mob-gamma-002 $\sqrt{\pi}$
 340. mob-gamma-003 $\sqrt{\pi}/2$
 341. mob-gamma-004 $3\sqrt{\pi}/4$
 342. mob-gamma-005 120
 343. mob-gamma-006 2
 344. mob-gamma-007 $\sqrt{\pi}/2$
 345. mob-gamma-008 6
 346. mob-gamma-009 $15\sqrt{\pi}/8$
 347. mob-gamma-010 1
 348. mob-wallis-001 $\pi/4$
 349. mob-wallis-002 $2/3$
 350. mob-wallis-003 $3\pi/16$
 351. mob-wallis-004 $8/15$
 352. mob-wallis-005 $\pi/4$
 353. mob-wallis-006 $3\pi/16$
 354. mob-wallis-007 $\pi/16$
 355. mob-wallis-008 1
 356. mob-wallis-009 1
 357. mob-wallis-010 $1/2$
 358. ser-001 收斂

359. ser-002 發散
360. ser-003 $3/2$
361. ser-004 條件收斂
362. ser-005 發散
363. ser-006 收斂
364. ser-007 $\exp(3)-1$
365. ser-008 0
366. ser-009 1
367. ser-010 條件收斂
368. ser-011 $9/4$
369. ser-012 1
370. ser-013 發散
371. ser-014 2
372. ser-015 6
373. ser-016 收斂
374. ser-017 3
375. ser-018 絕對收斂
376. ser-019 $3/4$
377. ser-020 發散
378. td-ser-001 $33/40$
379. td-ser-002 $13/12$
380. td-ser-003 $2*\log(2)$
381. td-ser-004 $3/2$
382. ser-021 1
383. ser-022 2
384. ser-023 條件收斂
385. ser-024 發散
386. ser-025 收斂
387. ser-026 3
388. ser-027 $2/3$
389. ser-028 收斂
390. ser-029 $\exp(3)-1$
391. ser-030 5
392. ser-031 發散
393. ser-032 發散
394. ser-033 條件收斂
395. ser-034 0
396. ser-035 Infinity
397. ser-036 收斂
398. ser-037 收斂

399. ser-038 $1/3$
400. ser-039 $1/2$
401. ser-040 6
402. gap-ser-root-001 $1/3$
403. gap-ser-root-002 1
404. gap-ser-lc-001 divergent
405. gap-ser-lc-002 convergent
406. gap-ser-cond-001 conditional
407. gap-ser-end-001 conditional
408. gap-ser-end-002 divergent
409. gap-ser-taylor-001 $1/120$
410. gap-ser-taylor-002 $1-x^2/2+x^4/24$
411. gap-tech-005 root test
412. mob-tech-007 ratio test
413. mob-tech-008 root test
414. mob-tech-019 endpoint analysis
415. mob-trap-004 conditional
416. mob-trap-005 check endpoints
417. mob-conv-001 divergent
418. mob-conv-002 convergent
419. mob-conv-003 conditional
420. mob-conv-004 absolute
421. mob-conv-005 convergent
422. mob-conv-006 convergent
423. mob-conv-007 divergent
424. mob-conv-008 divergent
425. mob-conv-009 convergent
426. mob-conv-010 divergent